

EVANSTON, WY, USA

WMO: 725775

Lat: 41.273N

Lon: 111.031W

Elev: 2183

StdP: 77.70

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 04111

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.2	-16.2	-22.4	0.7	-15.6	-19.9	0.8	-13.1	12.8	-2.3	11.8	-2.2	3.6	230	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	30.3	13.0	28.9	12.5	27.5	12.2	14.8	25.0	14.0	24.8	13.4	24.2	6.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	11.3	16.6	10.5	10.3	16.3	9.1	9.4	16.5	48.3	24.8	46.1	24.5	44.0	23.9	17.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.4	11.0	9.5	DB	-22.3	32.3	2.8	1.1	-24.3	33.1	-25.9	33.7	-27.5	34.3	-29.5	35.1
			WB	-22.7	16.0	2.7	0.9	-24.7	16.6	-26.2	17.1	-27.8	17.6	-29.8	18.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-5.5	-4.6	-0.2	4.3	9.3	15.1	20.3	18.8	13.9	6.8	0.1	-5.6		
	DBStd	10.15	4.91	4.87	4.87	4.65	4.74	4.41	2.47	2.70	4.37	5.04	5.82	5.14		
	HDD10.0	2386	480	410	318	177	72	12	0	0	18	121	296	483		
	HDD18.3	4588	738	643	575	421	279	111	11	25	140	358	546	742		
	CDD10.0	973	0	0	1	6	51	165	319	274	134	22	1	0		
	CDD18.3	133	0	0	0	0	1	14	72	41	6	0	0	0		
	CDH23.3	1851	0	0	0	0	13	254	908	559	114	2	0	0		
	CDH26.7	471	0	0	0	0	1	54	286	119	11	0	0	0		
(14) Wind	WSAvg	4.4	4.4	4.7	4.6	4.9	4.5	4.5	4.0	4.1	4.2	4.4	4.3	4.4		
(15) Precipitation	PrecAvg	409	41	32	34	43	49	32	18	23	30	37	33	37		
	PrecMax	584	77	63	73	98	126	107	48	57	88	106	62	94		
	PrecMin	268	14	10	12	14	12	3	1	3	6	1	7	5		
	PrecStd	80	21	13	16	19	29	25	12	14	19	21	15	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	9.0	16.2	21.0	25.0	30.5	32.2	30.9	27.9	22.6	16.3	7.8		
		MCWB	2.0	1.9	5.3	7.9	10.8	13.0	13.9	13.1	11.6	8.7	5.5	1.8		
	2%	DB	4.1	5.5	12.5	17.6	22.7	28.2	31.0	29.1	26.2	19.7	13.7	4.8		
		MCWB	0.2	0.9	4.1	6.0	9.7	11.7	13.3	12.4	10.9	7.5	4.5	0.7		
	5%	DB	2.5	3.7	9.5	15.4	21.0	26.3	29.7	27.9	24.6	17.6	10.7	2.8		
		MCWB	-0.5	0.2	2.8	5.3	9.0	11.3	13.0	12.2	10.1	6.7	3.1	-0.4		
	10%	DB	1.1	2.2	7.1	12.8	18.4	24.3	28.3	26.8	22.7	15.2	8.2	1.2		
		MCWB	-1.4	-0.8	1.7	4.3	7.9	10.8	12.7	11.9	9.5	5.7	2.2	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.9	6.0	8.4	11.9	14.2	15.8	15.2	13.9	10.2	6.7	2.8		
		MCDB	5.5	7.2	14.4	20.0	23.0	27.1	25.2	24.7	22.9	17.6	14.0	6.2		
	2%	WB	1.1	1.7	4.4	6.8	10.5	13.0	15.1	14.4	12.5	8.7	5.0	1.3		
		MCDB	3.2	4.5	11.7	15.8	20.5	25.1	25.1	24.4	21.6	16.8	11.4	3.8		
	5%	WB	0.0	0.6	3.2	5.6	9.5	12.1	14.5	13.7	11.5	7.6	3.8	0.1		
		MCDB	2.0	3.3	9.0	14.0	18.9	23.6	25.1	23.9	20.8	15.7	10.0	2.4		
	10%	WB	-1.1	-0.5	1.8	4.7	8.5	11.4	13.8	12.9	10.4	6.4	2.6	-1.0		
		MCDB	0.7	1.9	6.6	12.1	17.3	22.5	24.8	23.3	19.9	14.0	7.7	1.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.0	9.2	11.5	12.3	14.2	14.9	14.8	14.0	12.0	10.0	7.8		
		MCDBR	8.8	8.5	12.2	15.1	15.1	16.3	16.4	16.2	15.8	14.8	13.2	8.9		
		MCWBR	6.4	5.6	6.6	7.3	6.8	6.8	6.0	6.1	6.5	7.0	7.0	6.0		
		MCDBR	7.8	7.6	11.6	14.0	13.8	14.8	14.1	14.3	13.7	13.5	12.0	8.2		
	5% WB	MCWBR	5.9	5.2	6.6	7.0	6.4	6.5	5.4	5.4	5.8	6.5	6.6	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.230	0.241	0.255	0.292	0.295	0.293	0.325	0.323	0.287	0.267	0.239	0.227		
	taud		2.424	2.423	2.498	2.440	2.514	2.559	2.478	2.472	2.568	2.597	2.578	2.494		
	Ebn at Noon		954	991	1008	981	980	978	942	937	959	950	942	934		
	Edh at Noon		76	90	95	109	104	100	107	104	88	75	65	65		
(44) All-Sky Solar Radiation	RadAvg		2.13	3.01	4.38	5.52	6.45	7.72	7.38	6.42	5.29	3.66	2.37	1.75		
	RadStd		0.18	0.26	0.31	0.32	0.56	0.55	0.37	0.30	0.37	0.26	0.25	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page